

18.06 Linear Algebra - Lectures 17, 18, 19
Textbook Problem Sets - Section 4.4, 5.1, 5.2

◦ Problem Set 4.4

1) a) ∴ Independent ; ~~Orthogonal~~ Orthogonal →

b) $Q = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ $Q^T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ $Q^T Q = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$
 ~~$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$~~

∴ Independent

1) a) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ∴ Independent ✓

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = (+1)(-1) + (0)(1) = -1 \neq 0$ ∴ Not orthogonal, not orthonormal ✓

b) $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$; $\begin{bmatrix} 0.4 \\ -0.3 \end{bmatrix}$ → independent ✓

$\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.4 \\ -0.3 \end{bmatrix} = (0.6)(0.4) + (0.8)(-0.3) = (0.24) - (0.24) = 0$
∴ Orthogonal ✓

$\|v\| = \sqrt{(0.6)^2 + (0.8)^2} = 1$

$\|w\| = \sqrt{(0.4)^2 + (-0.3)^2} = 0.5 \neq 1$ ∴ Not orthonormal ✓

c) $\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$; $\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$ ∴ Independent ✓

$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \cdot \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} = (\cos \theta)(-\sin \theta) + (\sin \theta)(\cos \theta) = 0$ ∴ Orthogonal ✓

$\|v\| = \sqrt{(\cos^2 \theta) + (\sin^2 \theta)} = 1$

∴ Orthonormal ✓

$\|w\| = \sqrt{(-\sin \theta)^2 + (\cos^2 \theta)} = 1$

$$3) a) A = \begin{bmatrix} - & - & - \end{bmatrix}$$

$$\|A\| = \sqrt{(a_{11})^2 + (a_{12})^2 + (a_{13})^2} = 4$$

$$a_{11} \cdot a_{21} \cdot a_{31} + a_{21} \cdot a_{22} \cdot a_{23} + a_{31} \cdot a_{32} \cdot a_{33} = 0$$

If A orthonormal $\therefore A^T A = I$

$$(\frac{1}{4}A^T)(\frac{1}{4}A) = \frac{1}{16}I$$

$$\therefore 4B = A$$

$$\therefore A^T A = 16I \quad \checkmark$$

b) $A_1 = 1B \quad A_2 = 2C \quad A_3 = 3D$

~~$$[B] [2C] [3D]$$~~
$$[B \quad 2C \quad 3D]$$

$$q_i^T q_j = 0 \quad (\text{when } q_i \text{ orthonormal})$$

$$q_i^T q_i = 1 \quad \therefore A = \begin{bmatrix} \boxed{1} & \boxed{2} & \boxed{3} \\ \vdots & \vdots & \vdots \\ \textcircled{1} & \textcircled{2} & \textcircled{3} \end{bmatrix}$$

$$A^T A \text{ orthonormal} = A^T A = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A^T A = \begin{bmatrix} (1)^2 & 0 & 0 \\ 0 & (2)^2 & 0 \\ 0 & 0 & (3)^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix} \quad \checkmark$$

6) $Q_1, Q_2 \rightarrow$ orthogonal matrices

$$(Q^T Q = I)$$

$$Q_1 \cdot Q_2 =$$

$$= Q_1 \cdot Q_2 I_1 \cdot I_2$$

$$= Q_1 (Q_1^T Q_1) \cdot Q_2 (Q_2^T Q_2)$$

$$\cancel{A} A(A^T A)^{-1} A^T$$

$$Q^T Q = I :$$

$$\Rightarrow (Q_1 Q_2)^T (Q_1 Q_2)$$

$$\Rightarrow Q_1^T Q_2^T Q_1 Q_2$$

$$\Rightarrow Q_1^T Q_1 \cdot Q_2^T Q_2$$

$$\Rightarrow (Q_1^T Q_1) (Q_2^T Q_2)$$

$$\Rightarrow (I)(I) = I$$

$\therefore Q_1 Q_2$ is orthogonal. ✓

9 a) $P = Q Q^T$ $q_1 = \begin{pmatrix} 0.8 \\ 0.6 \\ 0 \end{pmatrix}$ $q_2 = \begin{pmatrix} -0.6 \\ 0.8 \\ 0 \end{pmatrix}$ $P^2 = P$

$$P = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.6 & 0 \\ -0.6 & 0.8 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \checkmark$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} 3 \times 2 \\ 2 \times 3 \end{matrix} \quad P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = P \checkmark$$

b) $Q^T Q = I$

$$\cancel{Q Q^T}^2 (Q Q^T)^2$$

$$= Q Q^T Q Q^T$$

$$= Q (I) Q^T$$

$$= Q Q^T \checkmark$$

$$13) q_1 = \frac{a}{\|a\|} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$B = b - (a^T b) q_1$$

$$\begin{aligned} &= \begin{bmatrix} 4 \\ 0 \end{bmatrix} - \left(\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right) \left(\frac{1}{\sqrt{2}} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 0 \end{bmatrix} - \left(\frac{4}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -2 \end{bmatrix} \end{aligned}$$

$$\text{multiple} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \checkmark$$

$$16) q_1 = \frac{a}{\|a\|} = \frac{(4, 5, 2, 2)}{\sqrt{4^2+5^2+2^2+2^2}} = \frac{(4, 5, 2, 2)}{\sqrt{16+25+8}} = \frac{a}{\sqrt{49}} \\ = \frac{1}{7} (4, 5, 2, 2) \checkmark$$

$$q_2 = \frac{b}{\|b\|} = \frac{b}{\sqrt{1^2+2^2+0+0}} = \frac{1}{\sqrt{5}} (1, 2, 0, 0) \times$$

multiple of a closest to b? $\rightarrow \frac{1}{7} \times$

$$\begin{aligned} p &= \left(\frac{a^T b}{a^T a} \right) a = a^T b (a^T a)^{-1} a \\ &= \frac{\begin{pmatrix} 4 & 5 & 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}}{\begin{pmatrix} 4 & 5 & 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix}} a = \frac{14}{49} a = \frac{2a}{7} \end{aligned}$$

$$\text{multiple} = \frac{2}{7} \checkmark$$

$$q_2 = \frac{B}{\|B\|} ; B = b - (q_1^T b) q_1 = b - p$$

$$\begin{aligned} &= \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} - \frac{2}{7} \begin{bmatrix} 4 \\ 5 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{8}{7} \\ \frac{10}{7} \\ \frac{4}{7} \\ \frac{4}{7} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{1}{7} \\ \frac{10}{7} \\ -\frac{4}{7} \\ -\frac{4}{7} \end{bmatrix} \end{aligned}$$

$$\|B\| = \sqrt{\left(\frac{1}{7}\right)^2 + \left(\frac{10}{7}\right)^2 + \left(-\frac{4}{7}\right)^2 + \left(-\frac{4}{7}\right)^2} = \sqrt{\left(\frac{1}{49}\right) + \left(\frac{100}{49}\right) + \left(\frac{16}{49}\right) + \left(\frac{16}{49}\right)} = \sqrt{\frac{133}{49}} \\ = 1$$

$$\therefore q_2 = B = \frac{1}{7} \begin{bmatrix} -1 \\ 10 \\ -4 \\ -4 \end{bmatrix} \checkmark$$

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19) $A = QR$ then

$$A^T A = R^T R$$

= lower triangular times upper triangular

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 9 & 9 \\ 9 & 18 \end{bmatrix} = R^T R = \begin{bmatrix} 3 & 0 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

$$A = QR$$

$$Q = \frac{1}{\|A\|} A R^{-1}$$

$$Q = \frac{1}{9} \begin{bmatrix} -1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1/3 & -1/3 \\ 0 & 1/3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 2 & 4 \end{bmatrix} = Q \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & 4/3 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

$$Q = \frac{1}{3} \begin{bmatrix} -1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\begin{aligned} (-1)(3/9) &= -1/3 \\ (-1)(-3/9) + (1)(3) &= 1/3 + 3 = 10/3 \end{aligned}$$

$$\begin{aligned} (-1/3)(3) + 0 &= -1 \\ (2/3)(3) &= 2 \end{aligned}$$

$$\begin{aligned} (-1) + 2(3) &= 5 \\ (2) + 4(3) &= 14 \end{aligned}$$

22) $A = a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$q_1 = \frac{A}{\|A\|} = \frac{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}{\sqrt{5}}$$

$$\begin{aligned} B &= b - (q_1^T b) q_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(\left(\frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \left(\frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(\frac{1}{\sqrt{5}} \right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \\ &= \begin{bmatrix} 1 - 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix} \end{aligned}$$

$$q_2 = \frac{B}{\|B\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 - 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

$$(1)(1) + (1)(-1) + (2)(0) = 0$$

$$C = c - (q_1^T c) q_1 - (q_2^T c) q_2$$

$$C = C - (q_1^T C) q_1 - (q_2^T C) q_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \left(\left(\frac{1}{\sqrt{6}} \right) \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) \left(\frac{1}{\sqrt{6}} \right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} - (q_2^T C) q_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \left(\frac{1}{\sqrt{6}} (9) \right) \left(\frac{1}{\sqrt{6}} \right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \left(\left(\frac{1}{\sqrt{2}} \right) \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) \left(\frac{1}{\sqrt{2}} \right) \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{3}{2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} \times \rightarrow \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}$$

$$\hookrightarrow \left\{ 1 - 3/2 - 1/2 \quad 1 - 3/2 \quad 1 - 2 = -1 \right\}$$

$$26) C = C^* - (q_2^T C^*) q_2 \quad C^* = a_3 - (q_1^T a_3) q_1$$

$$C = \left[a_3 - (q_1^T a_3) q_1 \right] - \left(q_2^T \left[a_3 - (q_1^T a_3) q_1 \right] \right) q_2$$

$$= a_3 - (q_1^T a_3) q_1 - q_2^T a_3 q_2 + q_2^T (q_1^T a_3) q_1 q_2$$

$$= a_3 - (q_1^T a_3) q_1 - q_2^T a_3 q_2 + q_2^T q_1^T a_3 q_1 q_2$$

$$C = C - (q_1^T C) q_1 - (q_2^T C) q_2 \quad \times$$

$$\rightarrow C^* = C - (q_1^T C) q_1$$

$$C = C^* - (q_2^T C^*) q_2$$

$$\rightarrow C = C - (q_1^T C) q_1 - (q_2^T C^*) q_2$$

$$\therefore (q_2^T C^*) = q_2^T [C - (q_1^T C) q_1]$$

$$= q_2^T C - (q_1^T C) (q_2^T q_1)$$

$$= q_2^T C - (0)$$

$$\therefore C = C - (q_1^T C) q_1 - (q_2^T C) q_2 \quad \text{✓}$$

$q_2 \perp q_1$ by definition

$$\therefore q_2^T q_1 = 0$$

28) mn^2

~~$r_{ij} = q_i^T v_j$~~ $r_{ij} = q_i^T v_j$; $1 \leq j$
 m : dot product costs $\underline{m}$ (m entries)
 m : $v - v - r_{ij} q_i \rightarrow$ cost m (m entries)
 $\underline{2m}$
 per column j in A ($j-1$ ~~columns~~ ^{projections to subtract} per column)
 $\therefore$ total j 's $= \underline{2m(n-1)}$
 $= 2m(1 + \dots + (n-1))$
 $= 2m\left(\frac{n(n-1)}{2}\right)$
 $= mn(n-1)$
 $\{n-1 \approx n \text{ for large } n\}$
 $= mn(n) = \underline{mn^2}$
 Walkthrough of solution ~

30) $W = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & -1 & -\sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 & -\sqrt{2} \end{bmatrix}$

$W_1 \cdot W_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = (1) + (-1) + (-1) + (-1) = 0$ $W_1 \perp W_2$
 $W_2 \cdot W_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} \\ -\sqrt{2} \\ 0 \\ 0 \end{bmatrix} = (\sqrt{2}) + (-\sqrt{2}) + 0 + 0 = 0$ $W_2 \perp W_3$
 $W_3 \cdot W_4 = \begin{bmatrix} 0 \\ 0 \\ \sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ \sqrt{2} \end{bmatrix} = (0) + (0) + (0) + (0) = 0$ $W_3 \perp W_4$
 $W_1 \cdot W_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} \\ -\sqrt{2} \\ 0 \\ 0 \end{bmatrix} = 0$
 $W_1 \cdot W_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \sqrt{2} \\ -\sqrt{2} \end{bmatrix} = 0$
 $W_2 \cdot W_4 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \sqrt{2} \\ -\sqrt{2} \end{bmatrix} = 0$
 $W_3 \cdot W_2 = 0$
 $\rightarrow$ (check $W_i^T W_i = I$)
 all columns $\perp$ to each other. ortho normal

$W^T W = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} = 4 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$W^T W = D$

$\therefore W^{-1} = D^{-1} W^T$

$= \begin{bmatrix} 1/4 & 1/4 & 0 & 0 \\ 0 & 1/4 & 1/8 & 1/8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$
 $= \begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & -1/4 \\ \sqrt{2}/8 & -\sqrt{2}/8 & 0 & 0 \\ 0 & 0 & \sqrt{2}/8 & -\sqrt{2}/8 \end{bmatrix}$

◦ Problem Set 5.1

1) $\det A = \frac{1}{2}$

$$\det(2A) = 2^4 \det(A) = 16(\frac{1}{2}) = 8 \quad \checkmark$$

$$\det(-A) = (-1)^4 \det(A) = (+1)(\frac{1}{2}) = \frac{1}{2} \quad \checkmark$$

$$\det(A^2) = \det(A)\det(A) = (\frac{1}{2})^2 = \frac{1}{4} \quad \checkmark$$

$$\det(A^{-1}) = \frac{1}{\det(A)} = 2. \quad \checkmark$$

4) J_3 : Exchange Row 1 and Row 3 $\checkmark$
 J_4 : ... Row 1 and Row 4 $\checkmark$
 + Row 2 and Row 3. $\checkmark$

7) $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

~~$$\begin{bmatrix} \cos \theta & -\sin \theta & \cos \theta & -\sin \theta \\ 0 & 0 & 0 & 0 \end{bmatrix}$$~~

~~$$\det Q = \cos^2 \theta + \sin^2 \theta = 1 \quad \checkmark$$~~

$$Q = \begin{bmatrix} (1-2\cos^2 \theta) & (-2\cos \theta \sin \theta) \\ (-2\cos \theta \sin \theta) & (1-2\sin^2 \theta) \end{bmatrix}$$

$$\begin{aligned} \det Q &= [(1-2\cos^2 \theta)(1-2\sin^2 \theta)] - [(-2\cos \theta \sin \theta)(-2\cos \theta \sin \theta)] \\ &= [1 - 4\cos^2 \theta \sin^2 \theta + 4\cos^2 \theta \sin^2 \theta] - [4\cos^2 \theta \sin^2 \theta] \\ &= 1 - 4\cos^2 \theta \sin^2 \theta \end{aligned}$$

$$= [1 - 2\sin^2 \theta - 2\cos^2 \theta + 4\cos^2 \theta \sin^2 \theta] - [4\cos^2 \theta \sin^2 \theta]$$

$$= 1 - 2\sin^2 \theta - 2\cos^2 \theta$$

$$= 1 - 2(\sin^2 \theta + \cos^2 \theta)$$

$$= 1 - 2 = \underline{\underline{-1}} \quad \checkmark$$

$$10) A\alpha = 0 \quad \det A = 0$$

$$\sum A_{ii} = 0 \quad \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Rule 3: } \det A = 3(1) + 3(1) + 3(-2)$$

$$3\det A = \text{X}$$

→ Every row sum to zero $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ in $N(A)$.
 $\therefore A$ is singular

$\therefore$ Rule 8 $\det A = 0$ if A singular

Every row sum to 1
 $\hookrightarrow \{ \therefore \text{rows of } (A-I) \text{ sum to zero} \}$
 $\therefore \det A$ not necessarily 0

$$13) A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = U$$

$$\therefore \det(A) = (1)(1)(1) = 1 \checkmark$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$\det(A) = (1)(1)(2) = 2 \text{ x } 3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\det(A) = (1)(1)(3) = 3 \checkmark$$

17) $A^T = A^{-1}$ $\checkmark$ Not $A^{-1} \Rightarrow -1 \cdot A$

$$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & a & b+a \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & a & b+c \\ -a-b & -c & c \\ -b & -c & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a-b & a-c & b+c \\ -a & -b & c \\ -b & -c & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a-b & a-c & b+c \\ -a & -b & c \\ 0 & 0 & 0 \end{bmatrix}$$

$$A^T = A^{-1} = \begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$$

$\therefore \det(A^T) = \det(A^{-1})$

$\det(A^T) = \det(A)$ # Rule 10

$\det(A^{-1}) = \det(A)$

$\det(A^{-1}) = \frac{1}{\det(A)}$ # Rule 9

$\therefore \det(A) = \frac{1}{\det(A)}$

$\det(A)^2 = 1$ $\times$

$\det(K^T) = \det(-K) = (-1)^3 \det(K) = -\det(K)$

Rule 10 $\therefore \det(K^T) = \det(K)$

$\therefore \det(K) = -\det(K)$

$\therefore \det(K) = 0$

20) $\begin{bmatrix} a-Lc & b-Ld \\ c-La & d-Lb \end{bmatrix} = A$

$\det(A) = (a-Lc)(d-Lb) - (b-Ld)(c-La)$

$= ad - Lab - Lcd + LLbc - (bc - Lab - Lcd + LLad)$

$= ad - Lab - Lcd + LLbc - bc + Lab + Lcd - LLad$

$= ad - bc + LLbc - LLad$

$= (1-EL)ad - (EL-1)bc$

$= (ad-bc)(1-EL)$

$\therefore$ Different determinant if you do more than 1 row operation at once

$$23) A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\det(A) = (4)(3) - (1)(2) = \underline{10}$$

$$A^2 = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 18 & 7 \\ 14 & 11 \end{bmatrix}$$

$$\det(A^2) = \det(A) \det(A)$$

$$= 10^2$$

$$= \underline{100}$$

$$A^{-1} = \frac{1}{(4)(3) - (1)(2)} \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix}$$

$$\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{10}$$

$$A - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (4-\lambda)(3-\lambda) - (1)(2)$$

$$= 12 - 4\lambda - 3\lambda + \lambda^2 - 2$$

$$= \lambda^2 - 7\lambda + 10$$

$$= (\lambda - 2)(\lambda - 5)$$

$$\lambda = 2 \text{ or } \lambda = 5 \} \det(A - \lambda I) = 0$$

28) a) If A is not invertible then AB is not invertible.

↳ True ✓ ; $\det(A) = 0$ if not invertible

so $\det(A)\det(B) = 0$ always which means AB is singular and non-invertible.

~~False~~

b) Determinant of A is always the product of its pivots.

$$\text{False} \quad A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \Rightarrow U = \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}$$

$$(4)(5) = 20$$

$$\text{Row exchange} \Rightarrow -1 \times \det(A) \quad \det(A) = (4)(3) - (2)(1) = 10$$

$$10 \neq 20$$

28 c) The determinant of $A-B$ equals $\det A - \det B$

$$\hookrightarrow \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\det(A) - \det(B) = \cancel{\det(A-B)} \det(A-B)$$

$$\{10 - 1 = 9\} \neq \{(3)(2) - (1)(2) = 4\}$$

$$\det(A) - \det(B) \neq \det(A-B) \quad \checkmark$$

False $\checkmark$

d) AB and BA have the same determinant

$$\hookrightarrow \text{True } \checkmark \quad \det(AB) = \det(A) \det(B)$$

$$\det(BA) = \det(B) \det(A)$$

$$= \det(A) \det(B)$$

$$\therefore \det(AB) = \det(BA) \quad \checkmark$$